The Grain Is Not Given

Decoherence fixes the basis, not the decomposition: an empirical coda to the holism–disconnection convergence

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Abstract

Three companion papers argued that holism (*everything is related*) and disconnection (*nothing is related*) are empirical twins—each, in its strong reading, an idle lens—and earned a single positive datum: the world is **specifically relationally structured** (quantum non-separability shows it is neither a heap of intrinsic atoms nor a seamless whole). This coda presses that datum for its one unanswered question: *is the world's preferred decomposition—which wholes are real units, which relations irreducible—fixed by physics, or is it the modeller's frame?* The natural candidate for a physics answer is **einselection**: decoherence robustly selects a preferred *pointer basis*, so perhaps it also fixes the preferred *decomposition*. We test this directly. A pointer basis and a coarse decomposition are **distinct objects**: the basis is einselected by the system–environment coupling; the preferred coarse grain is the coarse-graining that maximises Hoel *effective information*—a **functional of the dynamics**, not of the coupling. In a small open quantum (Lindblad) system whose decoherence einselects an unambiguous basis, with the effective-information engine validated against a published emergent oracle, the effective-information-maximising grain is set by the dynamics and does **not** coincide with the einselected pointer sectors except by accidental alignment; the one published quantum causal-emergence example has the grain **anti-aligned** with the pointer basis (the quantum eraser maximises effective information); and the optimal grain is provably **non-unique**. The result is a **negative refinement**: decoherence fixes the *basis* of the world's relational structure but not its *decomposition*. The datum therefore has two layers—a basis that is the world's, and a grain that is partly the modeller's, dynamics-set and frame-relative—so the trilogy's lens verdict survives one level deeper: the world is relationally structured, but *which* relations count as the units is not simply read off it.

1 Introduction

1.1 The datum, and its one unanswered question

This is the fourth and last of a set. The first three took two opposite maximal theses to their strongest forms and found them empirical twins: holism, that every system is constitutively related, and disconnection, that nothing is related and the whole is an orphan, are—in their strong, ontological readings—empirically equivalent, each surviving only as a heuristic lens plus an unfalsifiable metaphysics. The convergence of the two earned exactly one fact about the world, bought with physics rather than metaphysics: quantum non-separability shows the world is **neither** a mere aggregate of intrinsic atoms (separability fails for entangled wholes) **nor** a seamless undifferentiated whole (entanglement is specific, local, and the exception against a mostly-separable background). The world is *relationally structured in a specific, measurable way* (Bell 1964; Healey and Gomes 2022; Ladyman and Ross 2007).

That datum leaves one question standing, and it is the sharpest one. To say the structure is

specific is to say some decomposition of the world—into which things are the units, which relations are the irreducible joints—is privileged. **Is that privileged decomposition fixed by physics, or is it the modeller's frame?** If physics hands it to us, the trilogy's deflationary verdict is bounded: the extremes are lenses, but the middle is the world's own structure, decomposition and all. If physics does *not* fix the decomposition, the lens reaches deeper than the trilogy showed: even the one positive datum would not tell us which relations count as the units.

1.2 The candidate physics answer, and why it must be tested

There is a natural candidate for a physics answer: **einselection**. Decoherence does not leave a system in an arbitrary state; the system–environment interaction robustly selects a preferred *pointer basis*, the basis whose states survive monitoring while their superpositions decay (Zurek 2003). Einselection is precisely physics fixing a preferred description—so it is the obvious mechanism by which the world might fix its preferred decomposition too. The conjecture under test (call it the *einselection-fixes-the-grain* claim) is that the coarse-graining a principled observer would call the system's real macro-structure coincides with the einselected pointer sectors.

This must be tested rather than assumed, because a pointer *basis* and a coarse *decomposition* are not the same object (section 2), and there is a published reason to doubt the alignment: the one existing quantum causal-emergence study finds effective information *maximised by the quantum eraser*—the coarse-graining that *destroys* which-path (pointer) information (Rossi and Souza 2019). So the honest prior is open, and the question is empirical.

1.3 The result, and the method

The result.

> Decoherence (einselection) fixes the preferred **basis** of the world's relational structure, but **not** the preferred **decomposition**. The preferred coarse grain—operationalised as the coarse-graining that maximises Hoel effective information—is a functional of the *dynamics*, not of the einselecting coupling; it coincides with the einselected pointer sectors only by accidental alignment, it can be *anti*-aligned with them, and it is non-unique. Therefore the "specific relational structure" the convergence earned has two layers: a basis that is the world's (physics-fixed), and a decomposition that is partly the modeller's (dynamics-set, frame-relative, non-unique). The lens survives one level deeper.

This is stated in the **necessary-only**, **deflationary** register the trilogy holds: the empirical claim is *measured on tractable open-system model classes* plus a validated engine plus published primaries; its generalisation to all physical decoherence is a labelled conjecture; and the metaphysical reading reclassifies, it does not refute.

The method. We make the basis/grain distinction precise (section 2), then test the einselection-fixes-the-grain claim in a small open quantum system, under a design contract built to be *able to fail and able to succeed*: the effective-information engine is first validated against a published emergent oracle; the model is required to exhibit genuine causal emergence (a doubly-stochastic model, which cannot, is the documented dead end); the pointer sectors and the effective-information-maximising grain are computed by independent procedures; the candidate coarse-grainings include ones that cross-cut the pointer sectors; and the whole is put to a context-clean adversarial fairness audit, which is reported including the overstatement it caught and corrected.

Roadmap. Section 2 separates the two notions and gives the test and its result. Section 3 draws the negative refinement, confirms it against the state of the art, gives the verdict, and states what the paper does not claim. The appendix is the reproducibility record.

2 Basis versus grain: two notions, and the test that separates them

2.1 Two distinct objects, routinely conflated

The einselection-fixes-the-grain claim quietly identifies two things that are not the same.

A **pointer basis** is a choice of *basis* for the system's Hilbert space: the set of states the environment monitors without disturbing, selected by the system–environment coupling via the commutativity / predictability-sieve criterion (Zurek 2003). It answers *which states are robust.*

A **coarse decomposition** (a grain) is a *partition* of the system's states into macro-units: which microstates are grouped into one effective whole. The principled operationalisation of "the privileged grain" we adopt is Hoel's: the coarse-graining that **maximises effective information** (EI)—the mutual information between a maximum-entropy intervention and its effect, decomposed as *determinism − degeneracy*—so that the macro description "beats" the micro (Hoel 2017; Hoel et al. 2013). It answers *which grouping carries the most causal structure.*

These are different questions, and—this is the load-bearing point—they have different *inputs.* The pointer basis is a functional of the *coupling* (what the environment measures). The EI-maximising grain is a functional of the *dynamics* (the system's effective transition law). Nothing in the formalism makes the coupling and the dynamics agree; einselection fixes a basis, and the question is whether that basis's sector structure is also the dynamics' privileged grain.

2.2 The test

We test the claim in a small open quantum system, under a design contract built to be *able to fail and able to succeed* (the full method and the validation are in section A).

The engine, validated. The EI implementation reproduces a published causal-emergence oracle—a deterministic 16-state cross-coupled-AND system whose ($A\wedge B$, $C\wedge D$) coarse-graining lifts EI from 1.62 to 2.00 bits (CE = +0.38)—to machine precision, and returns the required null on the controls (a doubly-stochastic transition matrix gives CE = 0 exactly). The first gate is that the ruler is correct.

Genuine emergence required. A doubly-stochastic effective dynamics exhibits *no* causal emergence at any scale, so its "EI-maximal grain" is meaningless; this is a documented dead end (an earlier toy died there). Pure-dephasing decoherence is exactly this unital, doubly-stochastic case. To make the test meaningful, the open system must be dissipative at finite temperature (non-normal jump operators, KMS detailed balance), giving a non-doubly-stochastic effective transition matrix with a non-uniform Gibbs stationary state—the regime where emergence is *possible.* The model is required to clear this gate (CE $>$ 0, stable above the noise floor) before any verdict.

The comparison. With a model that genuinely emerges, the einselected pointer sectors (from the coupling) and the EI-maximising grain (from the dynamics) are computed by independent procedures, and the EI-max grain is searched over *all* partitions—including ones that cross-cut the pointer sectors, so the claim can be refuted.

2.3 The result

The EI-maximising grain is set by the dynamics and is **not**, in general, the einselected pointer sectors. In the model's primary emergent regime the EI-max grain is a finer, determinism-preserving partition that matches *neither* the dynamical degeneracy classes *nor* any independently-set pointer partition—it cross-cuts them. Coincidence between the EI-max grain and the pointer sectors is *reachable*—we exhibit a contrived near-degenerate case in which it holds, so the test is not rigged toward refutation—but it is **not robust**: it survives only in the degenerate limit where no finer grouping exists, and dissolves under modest noise. There is no physical principle in the model forcing the coupling (which fixes the basis) to agree with the dynamics (which fixes the grain); when they are set independently, they disagree.

Two results from the state of the art make the disagreement principled rather than an artifact of one toy. First, the only published quantum causal-emergence study finds EI **maximised by the quantum eraser**: the which-path (pointer-distinguishing) coarse-graining gives $\mathrm{EI} = 0$, and erasing the pointer distinction maximises it (Rossi and Souza 2019)—the EI-max grain is there *anti*-aligned with the pointer basis. Second, even within the dynamics alone the EI-maximising coarse-graining is provably **non-unique** (Liu et al. 2024): there is no single grain the dynamics privileges, let alone one the coupling must match. (An adversarial, context-clean fairness audit confirmed the engine, the emergence, and this reading—and caught and corrected an earlier overstatement of the positive case; see section A.)

The conclusion of the test is therefore narrow and firm: **einselection fixes the pointer basis but not the coarse grain.** The grain the world's dynamics privilege—if any single one does—is not the thing decoherence hands us.

3 The refinement, and the verdict

3.1 The two layers of the datum

The test forces a refinement of the convergence's one positive datum. "The world is specifically relationally structured" turns out to have **two layers**, with different owners:

- a **basis layer** that is *the world's*: decoherence einselects a preferred basis, robustly and observer-independently (Zurek 2003); and quantum non-separability is a fact about that basis's states no description can wish away (Bell 1964; Healey and Gomes 2022). Here physics does fix the structure.
- a **decomposition layer** that is partly *the modeller's*: which microstates count as one effective whole—the grain—is set by the dynamics, not by the einselecting coupling; it is non-unique even given the dynamics (Liu et al. 2024); and it can run opposite to the pointer basis (Rossi and Souza 2019). Here physics does *not* hand us a privileged decomposition.

So the datum is real but does **not** include its own decomposition. The world commits us to a basis and to specific irreducible joint facts; it does not commit us to a parsing of itself into units. That parsing remains, in part, a frame the modeller brings—exactly the "lens" the trilogy diagnosed at the extremes, now found one level inside the surviving middle.

3.2 B4 confirmed; the surviving-core symmetry

This is the empirical confirmation of the trilogy's structural result that no *preferred decomposition* is guaranteed. The holonic paper earned content only *relative to a pre-registered partition*; the

disconnection paper's separability was a property of a declared decomposition; the convergence found both reduce to a *(system, decomposition, signature)* tuple. The natural hope was that physics—decoherence— would supply the decomposition and dissolve the relativity. It does not: it supplies the basis and leaves the grain to the dynamics, non-uniquely. The surviving core of all four papers is therefore one stance, now with an empirical anchor: **the carving is the modeller's; what physics fixes (the basis, the non-separable facts) is not a parsing into units.** Map $\neq$ territory holds even of the territory's one measured structure—at the level of its decomposition.

3.3 Verdict

Deflationary, and one level deeper. The einselection-fixes-the-grain claim is **refuted as a universal**: decoherence fixes the basis, not the decomposition. The positive conditional—that the basis and the grain coincide *when* the einselecting coupling and the system's dynamics happen to pick the same partition—is *not excluded*, but it is not robustly demonstrable and is not generic; it is a special alignment, not a law. The world's specific relational structure is genuine at the basis layer and frame-relative at the decomposition layer. The trilogy's verdict thus extends cleanly: not only are the all-related and nothing-related extremes lenses, but the one fact that survives between them still does not deliver a preferred decomposition—so the deflationary reading reaches the deepest layer the comparison can touch.

3.4 What this paper does not claim

- **Not that decoherence does nothing, or that there is no objective structure.** Einselection of the basis and quantum non-separability are objective and physics-fixed; the claim is precisely that these do *not* extend to fixing the decomposition.
- **Not that no decomposition is ever privileged.** Particular systems may have a dynamically dominant grain; the claim is that *einselection* does not fix it, that it is non-unique in general, and that it need not align with the pointer basis—so there is no *universal, physics-given* decomposition.
- **Not a proof about all physical decoherence.** The empirical claim is *measured* on tractable open-system model classes with a validated engine and two first-hand-confirmed published primaries; the generalisation is a labelled conjecture.
- **Not a refutation of the metaphysics.** Holism and disconnection remain coherent metaphysics, reclassified as idle in their strong readings, not falsified; this coda adds an empirical layer to that verdict, it does not re-open it.
- **No contested interpretation promoted to fact.** The basis-layer facts rest on V1 results; the Hoel-EI operationalisation of "the privileged grain" is a stated choice, not the only possible one; a different measure of "the real grain" could be explored, and the paper claims only that the *standard* causal-emergence measure gives this result.

3.5 The honest ceiling

Two limits bound the result, both already exercised in producing it. *First*, register: the refutation of the universal claim is *measured* (a validated engine, genuine emergence, two confirmed primaries); the two-layer metaphysical reading is an *argued* interpretation of that measurement, deflationary and labelled conjectural beyond the model classes. *Second*, provenance: there was no external human physicist to referee this (not viable here), so the corrector was

reality and computation plus a context-clean adversarial audit plus the author's standing veto—and that machinery earned its place, catching and forcing the correction of an overstated positive conditional during this very test. The refutation stands on the robust half; the positive conditional is left honestly open; and the invitation to an external expert reviewer remains the one check this apparatus cannot perform on itself.

A Reproducibility record

The test is a Tier-1 planned change: the design contract (DC1–DC8), the build strategies, and the verified method spec are the planning triad behind it, and the artifact is a single, runnable script.

Artifact. `docs/planning/convergence_paper/c4_sim/c4_open_system_test.py` (Python + NumPy/SciPy). Running it prints every gate result below.

Gates (all run first-hand).

- **DC6 — engine validated.** The effective-information code reproduces the published AND-gate emergent oracle (micro 1.6226, macro 2.0000 bits, CE = +0.3774, group sizes $[9, 3, 3, 1]$) to machine precision, and returns the null controls ($\mathrm{EI}(I_2) = 1$, $\mathrm{EI}(\text{uniform}) = 0$, any doubly-stochastic matrix $\Rightarrow$ CE = 0). The fairness audit independently re-derived the engine against the Kullback–Leibler form to $\sim 10^{-15}$.
- **DC1 — genuine emergence.** The thermal double-well (pure relaxation) does *not* emerge (CE < 0 across the barrier sweep) — an honest negative, and the reason a degeneracy mechanism is needed. The einselection-multiplicity model does emerge (CE $\approx +0.08$, stable across seeds above the noise floor).
- **P3 — comparison.** The EI-maximising grain $\neq$ the einselected pointer sectors in the primary emergent regime (it cross-cuts them); coincidence is reachable only in a contrived near-degenerate case that collapses under noise; prior-sensitivity reported.
- **DC7 — fairness audit.** A context-clean skeptic ran the script first-hand and returned "SOUND-REFUTED (the universal claim), with an overstated conditional tail" — which was then corrected in the artifact and in section 2.3/section 3.3. The audit verified the engine, the emergence, the physical fairness of the independence assumption, and the Rossi–Souza anti-alignment reading.

Planning + method. The why and the design contract: `docs/planning/C4_TEST_MASTER_PLAN.md` (§7 holds the verdict). The how: `C4_TEST_IMPLEMENTATION_STRATEGIES.md`. The first-hand-verified methods (Hoel effective information; the einselection / predictability-sieve recipe; the Lindblad $\to$ non-doubly-stochastic transition matrix guarantee; and the two arXiv bounds): `C4_METHOD_SPEC.md`.

What could not be verified. The two arXiv primaries' equation-level details (Liu et al. (2024) non-uniqueness form; Rossi and Souza (2019) eraser sweep) were read first-hand by a sub-agent and re-confirmed by the author at the level of each paper's existence and headline claim, not every equation. The metaphysical refinement (section 3) is an argued reading of the measured result, deflationary and conjectural beyond the simulated model classes.

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